WORKED SOLUTION

STEP II 2018, Question 13

Mechanics • Pass the parcel

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Read the question before you go on. It is in the STEP Support Programme's question database:

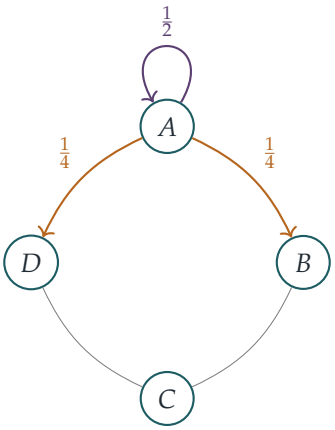
step.maths.org/questions/18-s2-q13

The key idea

Wherever the parcel is, the next whistle moves it by a fixed rule, so each of A_{n+1}, B_{n+1}, C_{n+1}, D_{n+1} is a fixed combination of A_n, B_n, C_n, D_n. Four coupled recurrences look daunting, but the right *combinations* (sums and differences of opposite children) each satisfy a one-line recurrence of their own, and those we can solve at once.

Solution

Label the children around the circle. On each whistle the holder keeps the parcel with probability $\frac{1}{2}$ and passes it to each neighbour with probability $\frac{1}{4}$; note that, by the question's set-up, clockwise and anticlockwise passes are equally likely.



The figure shows the moves from A; the same pattern applies at every child.

Part (i)

Starting from A, after one whistle

$$A_1 = \tfrac{1}{2}, \qquad B_1 = \tfrac{1}{4}, \qquad C_1 = 0, \qquad D_1 = \tfrac{1}{4},$$

since C is two places from A and cannot be reached in a single move.

To be at a given child after the second whistle, the parcel was either already there and stayed (probability $\frac{1}{2}$), or was at a neighbour and was passed the right way (probability $\frac{1}{4}$ from each neighbour). So

$$\begin{aligned} A_2 &= \tfrac{1}{2}A_1 + \tfrac{1}{4}B_1 + \tfrac{1}{4}D_1 = \tfrac{1}{4} + \tfrac{1}{16} + \tfrac{1}{16} = \tfrac{3}{8}, \\ B_2 &= \tfrac{1}{2}B_1 + \tfrac{1}{4}A_1 + \tfrac{1}{4}C_1 = \tfrac{1}{8} + \tfrac{1}{8} + 0 = \tfrac{1}{4}, \\ C_2 &= \tfrac{1}{2}C_1 + \tfrac{1}{4}B_1 + \tfrac{1}{4}D_1 = 0 + \tfrac{1}{16} + \tfrac{1}{16} = \tfrac{1}{8}, \\ D_2 &= \tfrac{1}{2}D_1 + \tfrac{1}{4}C_1 + \tfrac{1}{4}A_1 = \tfrac{1}{8} + 0 + \tfrac{1}{8} = \tfrac{1}{4}. \end{aligned}$$

As a check, $\frac{3}{8} + \frac{1}{4} + \frac{1}{8} + \frac{1}{4} = 1$.

Part (ii)

The argument used for $n = 2$ works for every $n \geq 0$, by the law of total probability (conditioning on who holds the parcel after the nth whistle):

$$\begin{aligned} A_{n+1} &= \tfrac{1}{2}A_n + \tfrac{1}{4}(B_n + D_n), & B_{n+1} &= \tfrac{1}{2}B_n + \tfrac{1}{4}(A_n + C_n), \\ C_{n+1} &= \tfrac{1}{2}C_n + \tfrac{1}{4}(B_n + D_n), & D_{n+1} &= \tfrac{1}{2}D_n + \tfrac{1}{4}(A_n + C_n), \end{aligned}$$

with $A_0 = 1$ and $B_0 = C_0 = D_0 = 0$.

Finding B_n and D_n. Adding the equations for B_{n+1} and D_{n+1},

$$B_{n+1} + D_{n+1} = \tfrac{1}{2}(B_n + D_n) + \tfrac{1}{2}(A_n + C_n) = \tfrac{1}{2}(A_n + B_n + C_n + D_n) = \tfrac{1}{2},$$

because someone always holds the parcel, so the four probabilities sum to 1.

> **Ask yourself** The sum $B + D$ is now known, but how do we split it between B and D?

Subtract instead of adding: the $A_n + C_n$ terms cancel, leaving

$$B_{n+1} - D_{n+1} = \tfrac{1}{2}(B_n - D_n).$$

Since $B_0 - D_0 = 0$, this gives $B_n - D_n = 0$ for every n. (This is the mirror symmetry of the game: passing clockwise and anticlockwise are equally likely, and B and D sit symmetrically either side of A.) Combining with $B_n + D_n = \frac{1}{2}$,

$$B_n = D_n = \tfrac{1}{4} \qquad \text{for all } n \geq 1,$$

while of course $B_0 = D_0 = 0$. This agrees with part (i).

Finding A_n and C_n. For $n \geq 1$ we now know $A_n + C_n = 1 - (B_n + D_n) = \frac{1}{2}$. Subtracting the equations for A_{n+1} and C_{n+1}, the $B_n + D_n$ terms cancel:

$$A_{n+1} - C_{n+1} = \tfrac{1}{2}(A_n - C_n),$$

and this holds for all $n \geq 0$. As $A_0 - C_0 = 1$, we get $A_n - C_n = \left(\frac{1}{2}\right)^n$ for all $n \geq 0$. Solving with $A_n + C_n = \frac{1}{2}$:

$$\boxed{A_n = \frac{1}{4} + \frac{1}{2^{n+1}}, \qquad C_n = \frac{1}{4} - \frac{1}{2^{n+1}} \qquad (n \geq 1).}$$

These formulas fail at $n = 0$ (they would give $A_0 = \frac{3}{4}$), because the relation $A_n + C_n = \frac{1}{2}$ only starts after the first whistle.

Checks

- $n = 1$: $A_1 = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$ and $C_1 = 0$; $n = 2$: $A_2 = \frac{3}{8}$ and $C_2 = \frac{1}{8}$. Both match part (i).
- $n = 3$: the formulas give $(A_3, B_3, C_3, D_3) = (\frac{5}{16}, \frac{1}{4}, \frac{3}{16}, \frac{1}{4})$, which sums to 1 and agrees with stepping the recurrences directly; a computer simulation of 200 000 games gave $A_3 \approx 0.313$ and $C_3 \approx 0.188$.
- As $n \to \infty$ every probability tends to $\frac{1}{4}$: after many whistles the parcel is equally likely to be anywhere, as symmetry suggests.

Common slips

- Writing B_{n+1} in terms of the probabilities of passing *from* B, rather than *into* B. Ask "where could the parcel have been just before?"
- Stating $A_n = \frac{1}{4} + 2^{-(n+1)}$ for all $n \geq 0$; it is wrong at $n = 0$.
- Assuming $B_n = D_n$ "by symmetry" without a word of reason; the difference recurrence makes it precise.

Beyond STEP

This is a *Markov chain*: a random walk on a cycle of four states with transition matrix P, and $(A_n, B_n, C_n, D_n) = (1, 0, 0, 0)P^n$. The combinations $B_n - D_n$, $A_n - C_n$ and $A_n + C_n - B_n - D_n$ come from the eigenvectors of P (with eigenvalues $\frac{1}{2}$, $\frac{1}{2}$ and 0), and the convergence to the uniform distribution at rate 2^{-n} is governed by the second-largest eigenvalue. Grimmett and Welsh, *Probability: An Introduction*, has a friendly treatment of Markov chains and their long-run behaviour.

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